Northward motion of the Whitehorse Trough: paleomagnetic evidence from the Upper Cretaceous Carmacks Group[1]

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The Upper Cretaceous Carmacks Group (70.4 ± 2.4 Ma) comprises gently dipping basaltic and andesitic lava flows overlying volcaniclastic deposits of the Intermontane Belt in the Whitehorse Trough. The sampling area is in southern Yukon and northern British Columbia; it lies west of the Tintina – Northern Rocky Mountain Trench fault and Teslin Suture Zone and east of the Denali–Shakwak fault. Volcanic sections were sampled in three regions spread over 300 km, providing the first paleomagnetic data from pre-Tertiary volcanic rocks in the northern Canadian Cordillera. Alternating-field and thermal demagnetization revealed stable magnetization for 18 of the 27 sites collected. The overall mean direction (D = 166.7°, I = −71.4°, k = 53, α_{95} = 4.8°, N = 18 sites) is pre-folding and is most probably primary (latest Cretaceous). This gives a paleopole at 109.4°E, 82.1°N, K = 21, A_{95} = 7.8°. A critical evaluation of North American cratonic data yields a reference paleopole for the latest Cretaceous at 185.8°E, 77.7°N, A_{95} = 7.7°, implying 13.4 ± 8.5° (1500 ± 950 km) northward displacement and 10.2 ± 20.7° (not significant) clockwise rotation of the Whitehorse Trough. The displacement occurred between 70.4 ± 2.4 and 59–54 Ma, the "docking" time suggested by other paleomagnetic studies in Alaska. The amount and timing of this displacement are consistent with Gabrielse's estimate of cumulative dextral displacements along the Northern Rocky Mountain Trench fault and associated lineaments. As expected, it is intermediate between the low paleolatitudes observed from middle Cretaceous rocks and the concordant paleolatitudes observed in Middle Eocene rocks of the Western Cordillera farther south, in British Columbia and in northern Washington. The estimated displacement is consistent with those predicted by kinematic models of Engebretson and Umhoefer based on North Pacific Basin plate motions.

Le groupe de Carmacks d'âge crétacé tardif (70,4 ± 2,4 Ma) comprend des coulées basaltiques et andésitiques légèrement inclinées, superposant les dépôts volcaniques de la zone d'Intermontane dans la fosse de Whitehorse. L'échantillonnage a été effectué dans le sud du Yukon et le nord de la Colombie-Britannique, à l'ouest de la faille Tintina – Montagnes Rocheuses du Nord et de la zone de suture de Teslin et à l'est de la faille Denali–Shakwak. Des échantillons ont été recueillis dans trois régions s'étendant sur plus de 300 km; il s'agit là des premiers résultats paléomagnétiques provenant de roches volcaniques prétertiaires de la Cordillère-Nord. Des désaimantations thermiques et par champs alternatifs ont révélé une aimantation stable pour 18 des 27 sites recueillis. L'aimantation moyenne (D = 166,7°, I = −71,4°, k = 53, α_{95} = 4,8°, N = 18 sites) a été acquise avant plissement et est fort probablement primaire (fin du Crétacé), donnant un paléopole à 109,4°E, 82,1°N, K = 21, A_{95} = 7,8°. Une analyse rigoureuse des données paléomagnétiques pour le craton Nord-Américain donne un pôle de référence pour la fin du Crétacé à 185,8°E, 77,7°N, A_{95} = 7,7°. Ceci implique un déplacement vers le nord de 13,4 ± 8,5° (1500 ± 950 km) et une rotation dans le sens horaire de 10,2 ± 20,7° (non significative) de la fosse de Whitehorse. Le déplacement s'est produit entre 70,4 ± 2,4 et 56–54 Ma, âge « d'accostage » suggéré par une autre étude paléomagnétique en Alaska. La distance et l'âge du déplacement sont en accord avec les estimations du déplacement dextre cumulatif le long de la faille des Montagnes Rocheuses du Nord et des failles qui lui sont associées, établies par Gabrielse. Il est également intermédiaire entre les basses paléolatitudes observées pour des roches d'âge crétacé moyen et les paléolatitudes coïncidantes avec le craton Nord-Américain observées pour des roches d'âge eocène moyen provenant de la Cordillère occidentale en Colombie-Britannique et dans l'État de Washington. Le déplacement obtenu paléomagnétiquement est en accord avec ceux prédits par les modèles d'Engebretson et d'Umhoefer basés sur les mouvements des plaques océaniques du bassin du Pacifique-Nord.

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Introduction

We have investigated the paleomagnetism of the Carmacks Group to test for northward (poleward) displacement and azimuthal rotation of the Whitehorse Trough (Intermontane Belt) relative to the North American craton since 70 Ma. The area in west-central Yukon and northwest British Columbia, lying east of the Denali–Shakwak fault and west of the Tintina – Northern Rocky Mountain Trench fault, encompasses the Yukon Crystalline and Cataclastic terranes, the Teslin Suture Zone (Tempelman-Kluit 1976, 1980), and the Whitehorse Trough (Wheeler 1961; Bultman 1979). The Whitehorse Trough overlies the northern part of the Intermontane Superterrane (Tempelman-Kluit 1976; Tipper *et al.* 1981), an important tectonic element of the Canadian Cordillera (Fig. 1). Scattered through the Whitehorse Trough are remnants of volcanic strata, which are termed collectively the Carmacks Group (70 Ma). Large-scale motion relative to North America of much of the western Cordillera during Late Cretaceous and

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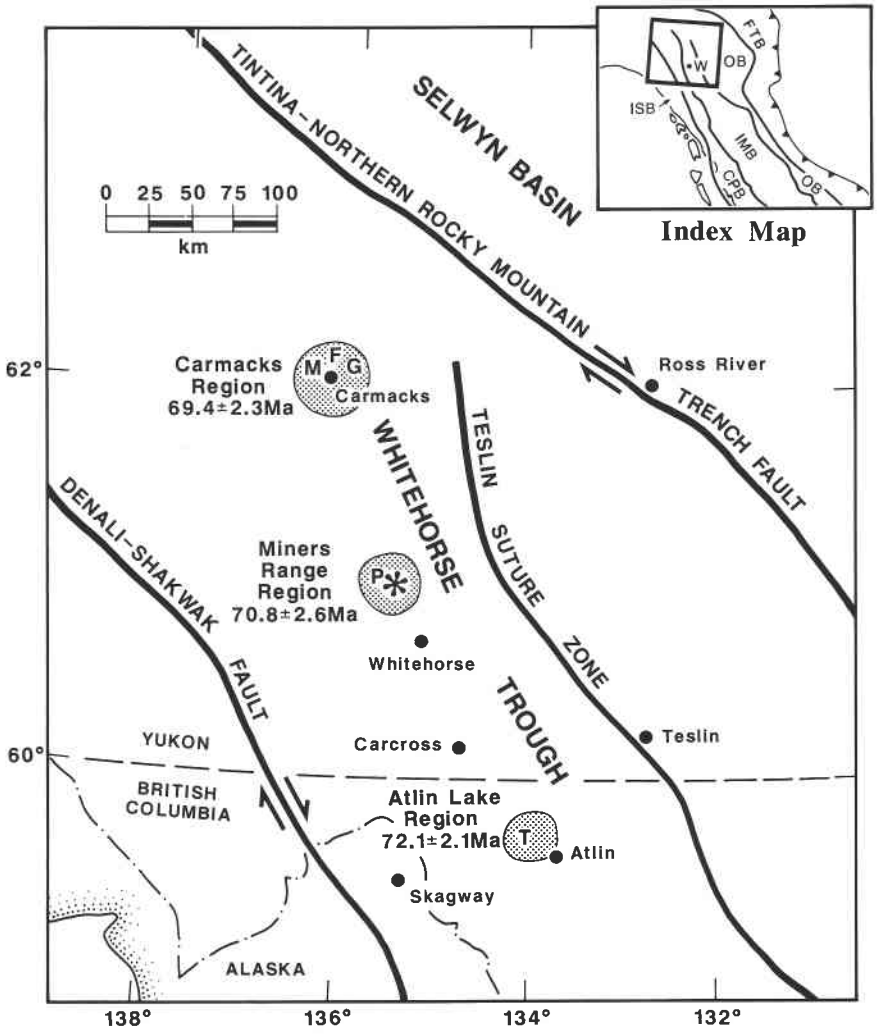


FIG. 1. Modified geologic map of west-central Yukon and northwest British Columbia showing locations and average ages of Carmacks, Miners Range, and Atlin Lake regions. Paleomagnetic sampling sections: Carmacks region—Miller's Ridge (M), Five Finger Mountain (F), Glenlyon (G); Miners Range region—Pilot Mountain (P); Atlin Lake region—Table Mountain (T). Average ages of regions determined in Table 1. Star denotes mean-site coordinates (135.5°W, 61.2°N). Estimated dextral displacements on faults: Tintina – Northern Rocky Montain Trench, 900 km (Late Cretaceous to Paleogene); Denali–Shakwak, 400 km (Tertiary). Geology modified from Tempelman-Kluit (1980) and Wheeler and McFeely (1987). Inset, main tectonic units of the Canadian Cordillera. FTB, Fold and Thrust Belt; OB, Omineca Belt; IMB, Intermontane Belt; CPB, Coast Plutonic Belt; ISB, Insular Belt.

early Tertiary time has been proposed on grounds of stratigraphic discontinuities and paleomagnetic data (Gabrielse 1985; Irving *et al.* 1985). The displaced region includes the Intermontane Superterrane, along with Wrangellia, Alexander, and other outboard terranes (Monger and Berg 1987; Wheeler and McFeely 1987). This large composite terrane is termed "Baja British Columbia" (Irving 1985). The paleomagnetic case is based on the divergence of paleopoles observed from Baja British Columbia with those observed from the craton. However, the main weakness in the paleomagnetic evidence is that many of the data come from intrusions for which there is little or no attitudinal control. Hence systematic local or regional tilting could explain the observed divergences (Irving 1979; Beck *et al.* 1981*a*). In addition, these intrusions, being mid-Cretaceous in age, preceded the hypothesized northward motion of Baja British Columbia and hence provide no information on the kinematics of the motion itself. Data from the well-bedded Late Cretaceous Carmacks Group thus could provide an intermediate point for the displacement history and hence provide an indirect check on the uncertainty regarding the attitudinal correction.

In this paleomagnetic study, initially suggested by Grond *et al.* (1984), we estimate the displacement and rotation of the Whitehorse Trough since latest Cretaceous time, compare our data with other paleomagnetic studies in the Cordillera, and speculate on how this inferred motion may be accommodated geologically and by plate-tectonic models.

Geology and sampling

Stratigraphic nomenclature

The Carmacks Group has had a somewhat confusing history of nomenclature and age assignment (Late Cretaceous to Pliocene), which we shall briefly review. In west-central Yukon, Upper Cretaceous to lower Cenozoic volcanic strata were called the Carmacks Group (Cairnes 1910; Bostock 1936; Tempelman-Kluit 1974, 1978, 1980), whereas in southern Yukon and northwest British Columbia similar rocks were

TABLE 1. Compilation of isotopic dates for Carmacks Group

Region	Rock type	Latitude (°N)	Longitude (°E)	Method	Age (Ma ± σ)
Carmacks	Alkalic basalt	62.07	223.59	K−Ar wr	73.1±2.5
Carmacks	Trachybasalt	62.07	224.05	K−Ar bi	68.0±2.2
Carmacks	Andesite	61.97	223.78	K−Ar wr	67.9±2.3
Carmacks average					69.4±2.3
Miners Range	Andesite	61.17	224.37	K−Ar wr	72.4±2.5
Miners Range	Andesite	61.15	224.38	K−Ar pl	69.1±2.6
Miners Range average					70.8±2.6
Atlin Lake	—	59.63	226.05	Rb−Sr isochron	72.4±2.1
Grand average (n = 6)					70.4±2.4

NOTES: Data from Carmacks, Miners Range, and Atlin Lake regions obtained by Grond *et al.* (1984), The University of British Columbia Geochronology Laboratory; Rb−Sr isochron obtained from rhyolite, andesite, trachybasalt, and alkalic basalt. Averaging procedure from Dalrymple and Lanphere (1969). σ, one standard deviation. K analyses were made by atomic absorption; Ar analyses, by isotope dilution. Rb and Sr were determined by X-ray fluorescence analysis of fused disks and pressed powder pellets. Sr isotopic analyses were made on a modified NBS-type mass spectrometer with automated digital data collection. Decay constants used were as follows: $\lambda_\beta = 4.962 \times 10^{-10}$ year^{-1} and $\lambda_\epsilon = 0.581 \times 10^{-10}$ year^{-1} for K; 1.42×10^{-11} year^{-1} for Rb, and 1.167×10^{-4} for the atomic ratio of ^{40}K/K (Steiger and Jäger 1977). wr, whole rock; bi, biotite; pl, plagioclase.

called the Hutshi Group (Wheeler 1961; Bultman 1979). Bostock (1936) correlated the Hutshi Group with volcanic rocks in the Carmacks district, which he named for Mount Nansen. Geochronological work by Grond *et al.* (1984) demonstrated that the Carmacks, Mount Nansen, and Hutshi groups were essentially contemporaneous. Further geochronology by Lowey *et al.* (1986) showed that volcanic rocks of the Carmacks Group in northwest Yukon are similar in age and affinity to the Carmacks, Mount Nansen, and Hutshi groups to the south (Grond *et al.* 1984). Furthermore, Tempelman-Kluit (1984) revised the stratigraphic nomenclature in Carmacks and Laberge map areas and restricted the term Mount Nansen Group to older (middle Cretaceous) volcanic rocks located in the eastern Whitehorse Trough. In order to simplify the nomenclature, Wheeler and McFeely (1987) applied the term Carmacks Group to all Upper Cretaceous volcanic strata in Yukon and northwest British Columbia.

Grond *et al.* (1984) obtained five K−Ar whole-rock and mineral dates for andesites and basalts collected from the Miners Range region and from Carmacks (Fig. 1). They also calculated a Rb−Sr isochron (using basalt, andesite, and rhyolite) for a sample of flow-banded rhyolite collected on Table Mountain west of Atlin, British Columbia (Fig. 1). The average of these six radiometric ages is 70.4 ± 2.4 Ma, i.e., Late Cretaceous (Table 1). Note that there appears to be a north to south progression of mean ages from the Carmacks to the Atlin Lake regions, although strictly speaking this spatial variation is not significant. These two studies confirm the latest Cretaceous (Maastrichtian) age of the Carmacks Group.

Sampling localities and procedures

We sampled from the Carmacks, Miners Range, and Atlin Lake regions, spread over about 320 km (Fig. 1). In these regions sufficient geochronologic and geologic information is available. The geology of the sampling localities is summarized in Table 2. In the Carmacks and Miners Range regions, the geology was compiled from our field observations, from various maps and reports (Bostock 1936; Campbell 1967; Tempelman-Kluit 1974, 1978, 1980, 1984), and from measured stratigraphic sections described in two unpublished theses (Churchill 1980; Grond 1980). In the Atlin region the geology was compiled from our field observations, from the report of Aitken (1959), and from an unpublished thesis (Bultman 1979).

We collected a total of 201 oriented cores from 27 sites. Fifteen sites were located on ridges at elevations of about 1800 m. The others were situated along Yukon highways 2 and 4, near Carmacks. All magnetic compass readings were checked with a sun compass or by distant sightings. Nine sites (71 cores) were omitted from further analysis, for reasons discussed later. Of these, seven were from the Glenlyon section, and the others were from the Pilot Mountain and Table Mountain sections.

Carmacks region

Samples were collected from three sections in the Carmacks region. The sites that yielded coherent data are listed in stratigraphic order in Table 2, oldest at the bottom. The high-angle, northwest-trending Braeburn fault separates the Glenlyon and Five Finger Mountain sections on the east from Miller's Ridge summit section on the west (Tempelman-Kluit 1984). The sequence comprises immature volcanic sandstone and epiclastic breccia intercalated with thin lava flows and siliceous tuff. These units are intruded by dikes and sills and are overlain by thicker, massive to columnar-jointed, plagioclase-porphyritic andesite. The andesite in turn is overlain by massive, olivine- and pyroxene-bearing basalt.

The Glenlyon section is completely fault bounded and is separated from the other sections by tightly folded Jura-Cretaceous strata. It comprises sheared, vesicular andesite intercalated with epiclastic breccia and immature sandstone. Alteration appears pervasive, probably as a result of the extensive fracturing. It is therefore not surprising that only one site yielded consistent paleomagnetic data.

Where observed, the Five Finger Mountain section consists of blocky jointed, porphyritic basalt lava flows. Because of very limited exposure, it was impossible to determine directly the bedding attitude. Instead, the bedding was extrapolated from mapped attitudes of the nearest basalt outcrops located west of Five Finger Mountain, on the west side of the Brae-

TABLE 2. Paleomagnetic data by sites

Site	Stratigraphic distance (m)	n	s	s'	Type	D_H (°), I_H (°)	D_B (°), I_B (°)	k	α_{95} (°)	Pole σ (°E), λ (°N)
					Miller's Ridge summit section, Carmacks region (basalt, 223.33°E, 62.17°N, flat, X)					
KCG-15	150	8	15	12	9A, 3B, 3C	179, −68	179, −68	136	4	48, 79
KCG-12	150	8	13	9	6A, 3B, 4C	173, −61	173, −61	19	12	59, 70
KCG-14	120	6	9	7	7A, 2C	165, −69	165, −69	186	4	91, 78
KCG-13	30	8	12	12	11A, 1B	168, −79	168, −79	57	6	189, 82
KCG-11	0	8	13	5	2A, 3B, 8C	175, −65	175, −65	46	11	57, 74
					Five Finger Mountain section, Carmacks region (basalt, 223.74°E, 62.16°N, 270/5°, Z)					
KCG-10	230	6	7	7	6A, 1B	206, −60	197, −62	392	3	8, 68
KCG-09	0	5	6	6	5A, 1B	139, −78	124, −74	540	3	156, 64
					Glenlyon section, Carmacks region (andesite, 224.05°E, 62.06°N, 50°/20°, Y)					
KCG-03	0	9	12	7	2A, 5B, 5C	121, −71	176, −68	63	8	58, 78
					Pilot Mountain section, Miners Range region (andesite, 224.50°E, 61.03°N, 341°/10°, Y)					
KCG-17	150	7	11	11	4A, 7B	60, 86	1, 79	56	6	227, 83
KCG-18	90	9	17	13	8A, 5B, 4C	40, 80	11, 73	18	10	338, 84
KCG-20	50	7	11	8	6A, 2B, 3C	136, −83	151, −73	42	9	140, 75
KCG-21	30	8	11	5	3A, 2B, 6C	106, −63	119, −57	30	14	131, 46
KCG-19	0	8	10	10	9A, 1B	6, −79	73, −86	98	5	210, 58
					Table Mountain section, Atlin Lake region (andesite, 226.08°E, 59.64°N, 240°/13°, Y)					
KCG-35	900	7	10	10	9A, 1B	182, −65	152, −69	102	5	124, 73
KCG-34	890	7	11	11	9A, 2B	189, −63	162, −69	19	11	110, 78
KCG-38	770	6	6	5	3A, 2B, 1C	211, −72	170, −81	69	9	212, 77
KCG-37	750	7	11	10	9A, 1B, 1C	195, −60	173, −67	294	3	73, 80
KCG-33	0	6	9	8	5A, 3B, 1C	199, −51	185, −60	16	14	34, 71
Grand average						178, −73	167, −71	53	5	109, 82

NOTES: Within each section, sites are listed in stratigraphic order. Approximate up-section distances are measured from the bottom. Sections themselves are listed in probable stratigraphic order. Bedding attitudes are given by bedding down-dip azimuth and dip. The reliability of bedding attitudes are coded as follows: X, good, attitude determined at the site; Y, reasonable, attitude determined from geologic maps and distant sightings; Z, poor, attitude inferred from sparse evidence on geologic maps. n, number of samples collected; s, number of specimens; s', number of specimens used in calculating site-mean directions. Type of magnetization: A, good end point, small θ_{63}; B, no obvious end point or large θ_{63}; C, no end point, large θ_{63} or more than two standard deviations from site-mean direction. $D°$, $I°$, declination and inclination of the mean direction with respect to present horizontal (H) and bedding (B); k, estimate of Fisher (1953) precision; α_{95}, half-angle of cone of 95% confidence; ϕ, λ, longitude and latitude of paleopole relative to bedding.

burn fault (Tempelman-Kluit 1984). This inference is probably correct because the gentle westward dip (5°) is consistent on both sides of the fault (D. J. Tempelman-Kluit, personal communication, 1987).

The Miller's Ridge summit section comprises massive, flat-lying basalt. Sampling covered about 150 m of stratigraphic section around the summit of Miller's Ridge.

Miners Range region, Pilot Mountain

The section consists of massive to columnar-jointed porphyritic andesite containing phenocrysts of plagioclase and hornblende. It was sampled about 300 m below the summit of Pilot Mountain (Tempelman-Kluit 1984). Prominent colour banding is occasionally seen. The section dips 10° to the northwest. This bedding attitude (Table 2) was obtained from orientations of subhorizontal joints and colour bands. The sites span about 150 m stratigraphically. A reversed to normal polarity change is observed between sites KCG-20 and KCG-18 about 50 and 90 m from the lowest sampling site (Table 2).

Atlin Lake region

We sampled along the lakeshore and around the 1520 m summit of Table Mountain, west of Atlin. The Table Mountain section comprises predominantly porphyritic andesite, with subordinant varicoloured basalt, dacite, and rhyolitic tuff. The bedding attitude (Table 2) is from Aitken (1959), who observed a 10−15° southwest dip of the volcanic strata from a distant ridge. We were unable to verify this estimate of bedding at the outcrop because contacts and flow boundaries are absent and subhorizontal joints show no consistent orientation. Our samples spanned about 900 m stratigraphically.

Remanent magnetization

Methods

Measurements were made on a Schonstedt SSM1 spinner magnetometer coupled with an IBM PC. Specimens were demagnetized by alternating field (AF) or thermally using, respectively, a Schonstedt GSD-5 tumbler−demagnetizer and a Schonstedt TSD-1 shielded furnace. For most of the sites,

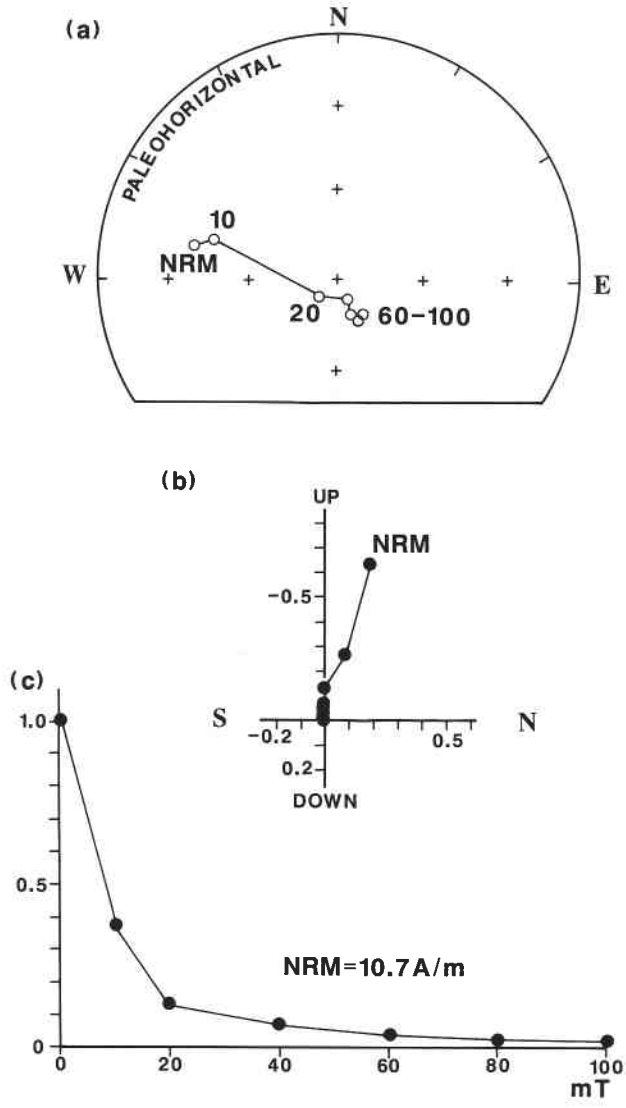


FIG. 2. AF demagnetization of A-type magnetization (site 20): (*a*) equal-area projection, (*b*) orthogonal plot, and (*c*) decay of normalized intensity of magnetization. In equal-area plots, solid (open) circle denotes positive (negative) inclination. In this example the original TRM is partially overprinted by an IRM probably caused by lightning.

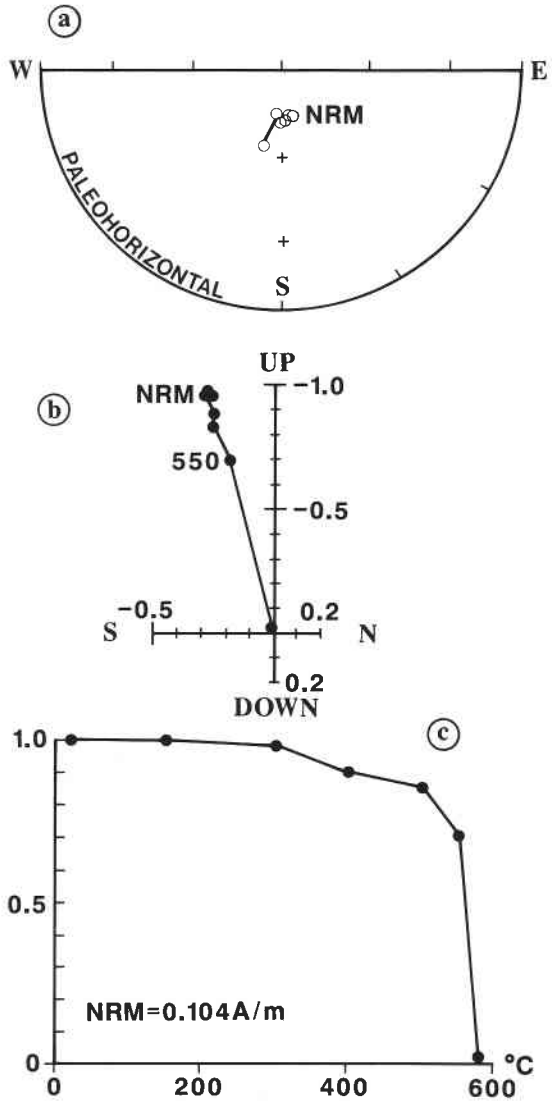


FIG. 3. Thermal demagnetization of A-type magnetization (site 34). Note the discrete unblocking temperatures indicated by the almost perfect magnetite "shoulder."

the optimum cleaning field was 40 mT, except for site 03 (20 mT), site 11 (30 mT), and sites 09, 10, and 18 (60 mT). The optimum cleaning temperature used was 550°C for most specimens.

Types of magnetization

The magnetizations of specimens have been grouped into three categories: A, B, and C. A-type magnetizations have stable end points, show good linear decay to the origin on orthogonal plots, and have values of $\theta_{63} \leq 7°$ (θ_{63} is the circular standard deviation of four measurements obtained on the magnetometer). B-type magnetizations have either acceptable end points but large θ_{63} (7–15°) or small θ_{63} but no defined end point. C-type magnetizations have no recognizable end points, have large values of θ_{63} ($> 15°$), or have directions of magnetization that are more than two angular standard deviations from the site-mean directions. Some of these aberrant magnetizations have weak intensities ($\sim 10^{-2}$ A/m) and have remanent directions almost parallel to the present Earth's field. Others have strong intensities (~ 10 A/m) and good within-specimen precisions ($\theta_{63} \sim 5°$C). However, neither AF nor thermal cleaning produced a change in direction, indicating that a "hard" isothermal remanent magnetization (IRM) has replaced the original thermoremanent magnetization (TRM). It seems probable that the IRM was induced by lightning, electrical storms being not infrequent in the region. This is supported by large (100–150°) magnetic-compass anomalies observed on Miller's Ridge. Because they contain no coherent paleomagnetic record, specimens with C-type magnetizations have been omitted from the analysis. Typical specimens and A- and B-type magnetizations are shown in Figs. 2–4.

A-type magnetization demagnetized by AF is shown in Fig. 2. Although there is a strong component with low coercivity (10 mT), probably caused by lightning, a stable, high-coercivity component is isolated by 60 mT. An example of ideal single-component A-type magnetization with a narrow range of unblocking temperatures (550–580°C) is shown in

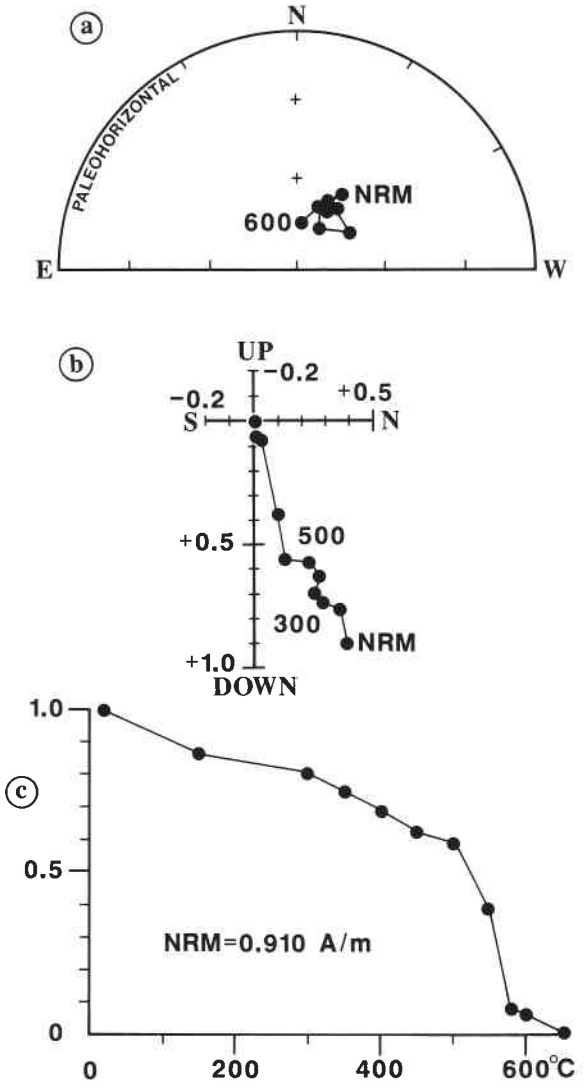


FIG. 4. Thermal demagnetization of B-type magnetization (site 17). Unblocking temperatures are more uniformly distributed than those in Fig. 3.

Fig. 3. This is characteristic of almost pure magnetite. In fact, in most of the specimens studied here, low-Ti titanomagnetite appears to be the main carrier of natural remanent magnetization (NRM).

An example of thermal demagnetization of a specimen with B-type magnetization is shown in Fig. 4. The specimen yields components with unblocking temperatures distributed from 300°C up to 500°C and with irregular directions, as shown by the orthogonal plot. However, above 500°C the decay is linear to the origin and the directions are well clustered.

Laboratory tests for magnetic properties

In order to determine the cause of this variable behaviour, certain laboratory experiments were carried out: IRM and low-field TRM acquisition and AF decay, and anisotropy of remanent magnetization. Typical IRM acquisition curves are shown in Fig. 5. The saturating fields range from 0.14 to 0.22 T. These specimens were then demagnetized by AF up to 100 mT (Fig. 6). In addition, specimens were given low-field TRM and then were demagnetized by AF. Normalized intensity-decay curves for NRM, IRM, and TRM are shown

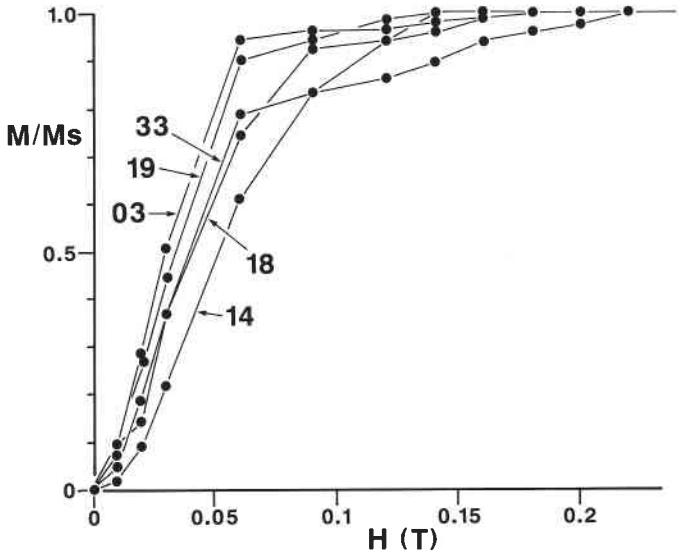


FIG. 5. Normalized IRM acquisition curves. M/M_s, ratio of IRM to saturation IRM; H, field. Saturation remanent magnetizations are as follows:

Site	Intensity (A/m)
03	184
14	372
18	221
19	122
33	49

in Fig. 6 for two specimens. The specimen from site 03 (andesite) exhibits typical multidomain grain behaviour because the saturation IRM is more stable than the low-field TRM (Lowrie and Fuller 1971). The NRM's for site 03 are very scattered because the original TRM has decayed rapidly, and the specimens then acquired soft, randomly distributed magnetizations. Most of the magnetizations from site 03 are B or C type (10 out of 12 specimens). The specimen from site 14 (basalt) exhibits typical single-domain grain behaviour, the low-field TRM being more stable than the saturation IRM. Except for two specimens with magnetizations that are directed toward the west and associated with lightning strikes, the NRM's from site 14 show a good cluster because the original TRM is very stable. Specimens from site 14 show mainly A-type magnetizations (7 out of 9 specimens). Therefore A- and C-type magnetizations are related to single-domain- and multi-domain-behaviour magnetic carriers, respectively. B-type magnetizations are probably related to magnetic carriers showing intermediate behaviour.

To test the possibility of inclination error due to preferred alignment of the grains during laminar flow of the lava, two experiments of IRM and TRM acquisition were made. Eight specimens were placed in a magnetic field of 0.25 T coaxial to the specimens. The magnetizations were measured before and after demagnetization in alternating fields of 25 mT. The companion specimens were heated to 700°C and then cooled in a magnetic field of 10^{-4} T coaxial to the specimens. Magnetizations were measured before and after thermal cleaning at 500°C. The observed inclinations before and after both AF and thermal demagnetization differed by no more than 0.9° from the direction of the applied field (Table 3). Hence

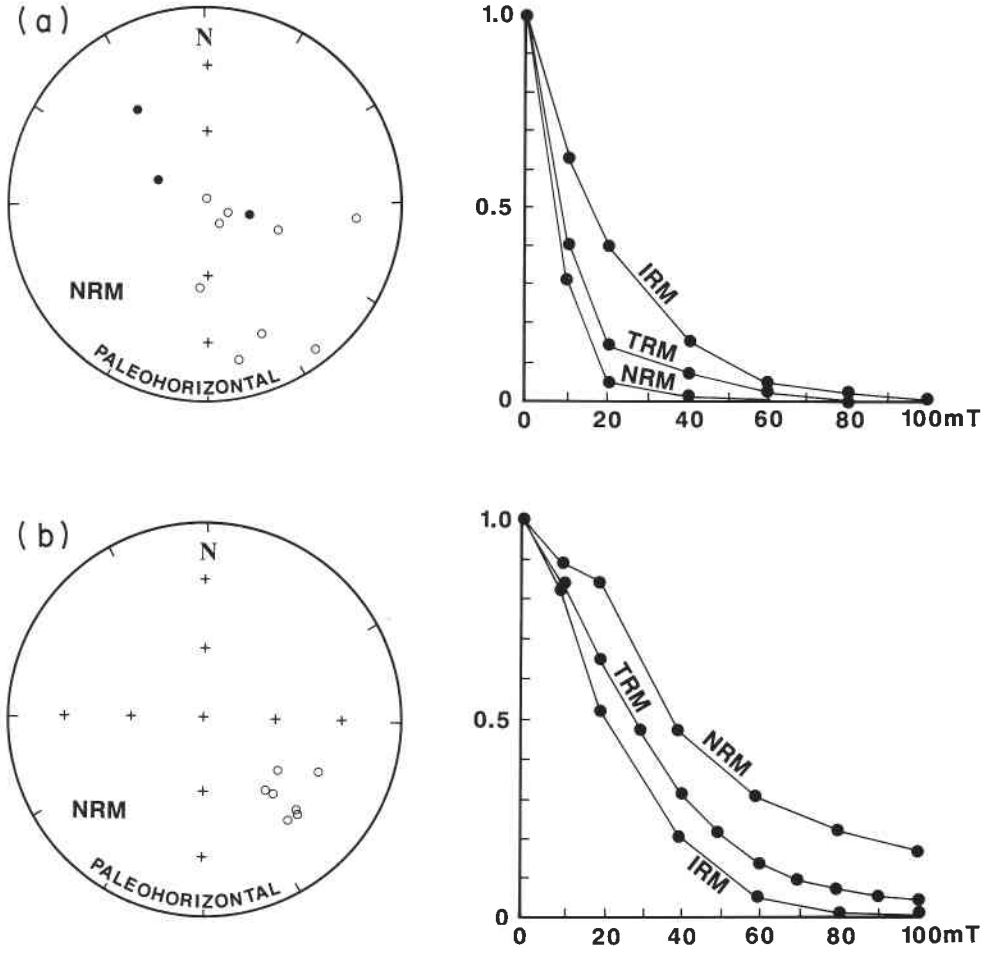


FIG. 6. Site-mean NRM's and comparison of AF demagnetization of NRM, TRM, and IRM for one specimen from each site. (*a*) Site 03, multidomain behaviour. (*b*) Site 14, single-domain behaviour.

it is improbable that the observed specimen directions of Table 2 are significantly affected by anisotropy.

Analysis

Mean direction and paleopole

Out of 27 sites sampled, 9 were omitted from calculation of the mean direction and paleopole. From the remaining 18 sites, 39 specimens were omitted. Directions of 156 specimens were used to calculate 18 site-mean directions (Fig. 7; Table 2). After the two normal-polarity sites are inverted through the origin, the mean direction, giving sites unit weight, is $D = 177.5°$, $I = -73.3°$, $\alpha_{95} = 6.7°$, with the Fisher (1953) precision parameter $k = 28$. After tilt correction the mean direction becomes $D = 166.7°$, $I = -71.4°$, $\alpha_{95} = 4.8°$, and $k = 53$. The increase in k obtained upon unfolding is significant; the data pass the fold tests of Watson (1956) and McFadden and Jones (1981) at more than 95% confidence. The paleopole obtained by averaging virtual geomagnetic poles (VGP's) calculated for each site is 109.4°E, 82.1°N, $A_{95} = 7.8°$, and $K = 21$.

Reversals and paleosecular variation

The mean directions of the normal group (2 sites) and reversed groups (16 sites) are nearly antipodal (172° apart). Because the Earth's magnetic field was of normal polarity about 50% of the time from 73 to 68 Ma (Harland *et al.* 1982), more than two normal-polarity sites would be expected if the sites had spanned several polarity intervals. There were, however, long periods of single polarity (up to 1.5 Ma) during this time interval. The fact that a polarity reversal is observed is an indication that the magnetization of the Carmacks Group probably spans sufficient time (at least 10^4 years) to average the effects of secular variation. This conclusion may be further tested by examining the angular dispersion (S_d) of the site-mean VGP's. The mean value of S_d is 17.7°, and the limits for this value, at 95% confidence, are 14.5 and 22.9° (Cox 1969). These limits show good concordance with the paleosecular variation model for the period 110–45 Ma (McFadden and McElhinny 1984). This agreement, combined with the presence of a polarity reversal, indicates that the data have

TABLE 3. Inclinations of synthetic magnetizations

	I (°)	α_{95}(°)	k
IRM (0.25 T)	89.7	1.8	949
IRM after 25 mT AF demagnetization	89.7	4.9	131
TRM (10^{-4} T)	89.5	3.3	285
TRM after 500°C demagnetization	89.1	5.1	112

NOTES: Averages of eight specimens. Applied field coaxial with cylindrical specimens.

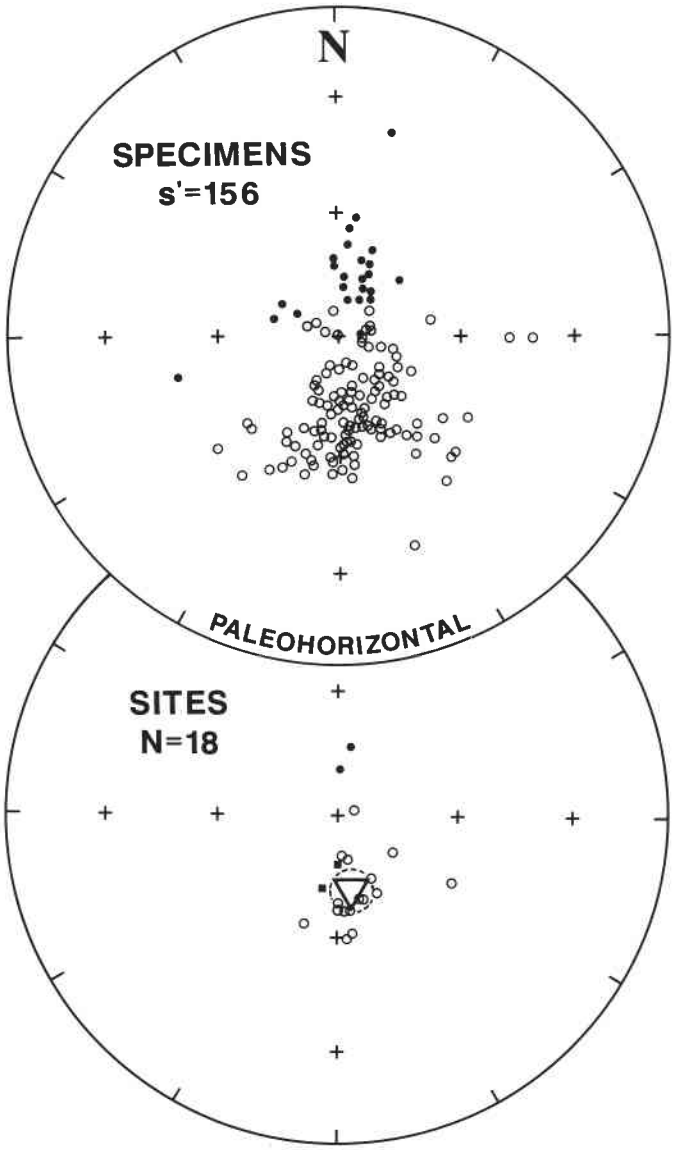


FIG. 7. Directions of stable remanent magnetization for the Carmacks Group, obtained by procedures described in text and in Table 3. Triangle denotes mean of 18 site-mean directions. Dashed circle is cone of 95% confidence. Squares indicate directions of normal-polarity sites inverted through the origin.

adequately averaged the paleosecular variation. Therefore, the mean paleopole obtained for the Carmacks Group very probably is representative of a geocentric axial dipole (GAD) field.

Timing of magnetization

There are two rock units younger than the Carmacks Group in the Whitehorse Trough—some Eocene volcanic centres scattered about Sloko, Bennett Lake, and Skukum, and the Pleistocene Selkirk Lavas—but neither of these units can be used to establish a younger limit on the timing of deformation. However, Tempelman-Kluit (1976) suggested that gentle tilting of the Carmacks Group occurred during or shortly after eruption. Furthermore, there is no evidence in the geochronology (Grond *et al.* 1984; Lowey *et al.* 1986) for a reheating event caused by later volcanism, plutonism, or deep burial. In summary, the available evidence indicates that the magnetization was acquired shortly after eruption in latest Cretaceous (Maastrichtian) time, about 70 Ma. Because the maximum coherence of site-mean directions results from complete rather than partial unfolding of the beds, it is unlikely that the magnetization coincided with deformation.

North American cratonic reference pole

To establish the North American cratonic reference pole for latest Cretaceous time, we decided to average several reliably dated paleopoles using a 15 Ma window centred at 70 Ma. We updated Irving and Irving's (1982) paleopole list as follows. We omitted the Late Cretaceous Elkhorn Volcanics and the Jefferson Valley sequence in Montana (Hanna 1967) because the ages of these rock units are not well constrained. As well we omitted the Niobara Formation of Kansas–Colorado–Wyoming because this rock unit is now considered early Late Cretaceous in age (Coniacian–Santonian) (Gordon *et al.* 1984), i.e., too old for this compilation. We added the 72.7 ± 4.2 Ma Roskruge Volcanics of Arizona (Vugteveen *et al.* 1981) and 35 Paleocene sites from the early Tertiary (64.3 ± 2.1 Ma) Alkalic intrusions of Montana (Diehl *et al.* 1983). Together with the Paleocene (63.3 ± 1.8 Ma) Gringo Gulch Volcanics of Arizona (Vugteveen *et al.* 1981), these selected paleopoles give a mean cratonic North American reference pole at 185.8°E, 77.7°N, A_{95} = 7.7°, K = 255 (Table 4).

Displacement and rotation of the Carmacks Group (Whitehorse Trough)

Comparison of paleopoles from the Carmacks Group and the North American craton suggests a 13.4 ± 8.5°northward displacement and a 10.2 ± 20.7° clockwise rotation of the Carmacks Group relative to the craton (Demarest 1983). The latter is not significant. Hence the paleomagnetic data indicate that the Whitehorse Trough has undergone some 1500 ± 950 km of northward displacement relative to the North American craton since 70 Ma (Fig. 8).

Discussion

Comparison with fault motions

This information is pertinent to tectonic reconstructions of the Canadian Cordillera. The Carmacks Group overlies portions of the Intermontane Belt, which is composed of Stikinia, Cache Creek, Quesnellia, and Slide Mountain terranes (Wheeler and McFeely 1987). The various elements of Baja British Columbia were amalgamated by Early Cretaceous time, prior to its accretion to the ancient margin of North America (Monger *et al.* 1982). The amount and timing of post-amalgamation northward translation of Baja British Columbia are important uncertainties in the tectonic evolution of the Cordillera.

The Whitehorse Trough is separated from the Selwyn Basin of the North American craton to the east by the Tintina – Northern Rocky Mountain Trench (TNRMT) fault, the Teslin Suture Zone (Fig. 1), and the Cassiar terrane. Hence, the post-70 Ma northward translation obtained above implies significant dextral motion on the TNRMT fault or the Teslin Suture Zone or both since latest Cretaceous time. A face-value interpretation of our displacement estimate (1500 ± 950 km) exceeds the minimum value of dextral offset on the Tintina Trench fault alone (450 km) proposed by Roddick (1967). It is consistent with the larger value of dextral offset of >900 km estimated by Gabrielse (1985) for the TNRMT fault and related lineaments. Timing of the motion on the faults described by Gabrielse ranges from mid–Late Cretaceous to perhaps Eocene, a time range also consistent with our time

TABLE 4. Cratonic poles for latest Cretaceous and Paleocene

Rock unit	Age (Ma)	Pole (°E, °N)	A_{95} (°)
Alkalic Intrusions, Montana (N = 35)	64.3±2.1[a]	181.4, 81.8	5.4
Gringo Gulch Volcanics, Arizona	63.3±1.8[b]	201.0, 77.0	1.5
Roskruge Volcanics, Arizona	72.7±4.2[b]	176.0, 73.6	7.7
Mean	64.6±2.9	185.8, 77.7[c]	7.7

[a]Diehl *et al.* 1983.
[b]Vugteveen *et al.* 1981.
[c]K = 255.

interval. The possiblity of post-70 Ma displacement along the Teslin Suture Zone is slight.

Comparison of displacements for plutonic and volcanic rock units

A second useful application of paleomagnetic data from the Carmacks Group is to compare estimates of the northward displacement inferred from bedded volcanic strata in the Whitehorse Trough with those from intrusive rocks in the Coast Plutonic Belt (CPB) of southwest British Columbia and northwest Washington. This is important because a number of paleomagnetic studies of middle Cretaceous intrusions in the CPB suggest large-scale (~25°) northward displacement (see Irving *et al.* 1985). The consistency of anomalously shallow paleolatitudes observed in several intrusions argues for displacement from the south, from about the present-day latitude of Baja California. However, Irving (1979) and Beck *et al.* (1981*a*) proposed an alternative hypothesis, that of post-emplacement tilting of the plutons about a uniform strike without poleward translation. This is because plutonic rocks generally lack recognizable indicators of bedding, so paleomagnetic directions from intrusions usually cannot be corrected for tilt. If anomalous paleolatitudes are observed, they may be ascribed incorrectly to translation rather than tilt. However, the present study indicates a northward translation of 13.4 ± 8.5° for the Carmacks Group, a stratified volcanic sequence for which bedding attitudes are known. It should be noted, however, that the Carmacks Group is about 20−30 Ma younger than the plutons studied paleomagnetically in the Coast Plutonic Complex. Hence the latter may have undergone some northward translation prior to eruption of the Carmacks Group. Therefore our data are consistent with the lower paleolatitudes observed in older intrusive rocks in the CPB and argue against the tilting hypothesis, although more paleomagnetic studies of cordilleran volcanic rock units will be needed to rule it out completely.

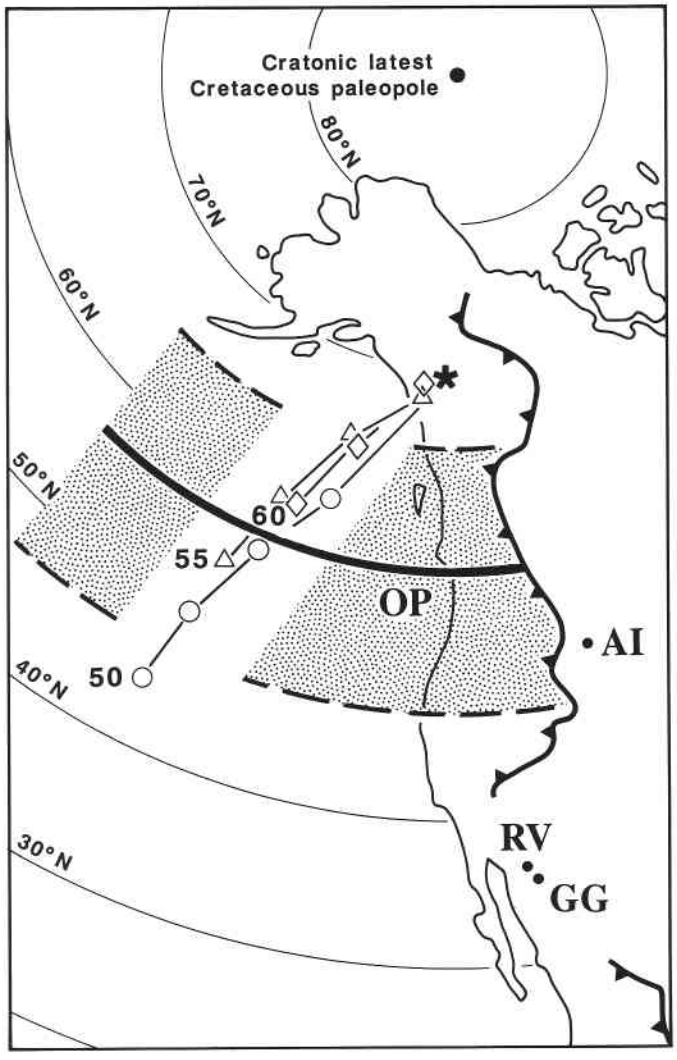


FIG. 8. Observed paleolatitude (OP) and error zone (stippled area) of the Carmacks Group. Lines connecting open symbols represent trajectories with 5 Ma intervals for the Whitehorse Trough and their related docking times (60, 55, and 50 Ma). Rotation poles used are from Engebretson *et al.* (1985). Circles denote rock units whose paleopoles were used to calculate the North American cratonic paleopole for latest Cretaceous time. Barbed line indicates approximate eastern limit of Cordilleran deformation. The error zone (P = 0.05) includes errors in the base map.

Integration of observed displacement into plate-tectonic models

The northward displacement could have been caused if these intrusions were attached to the Kula plate (Beck *et al.* 1981*b*; Debiche *et al.* 1987, Fig. 14C). In order to test the idea of rigid coupling between the Carmacks Group (and its basement) and the Kula plate, we used the Euler poles and rotation angles describing the motion of the Kula plate with respect to North America (Engebretson *et al.* 1985) and calculated possible trajectories (Kula trajectory models) for the Carmacks Group, assuming docking times of 60, 55, and 50 Ma (Fig. 8). The trajectories assume a perfect coupling with the Kula plate. The modelled latitude of formation for the Carmacks Group is in good accord with the latitude determined paleomagnetically if docking times of 60 or 55 Ma are assumed. If a later docking time of 50 Ma is postulated, the modelled paleolatitude is somewhat too shallow (Fig. 8). Furthermore, the trajectory models suggest significant clockwise rotations that could be accommodated by the large errors in the paleomagnetic data.

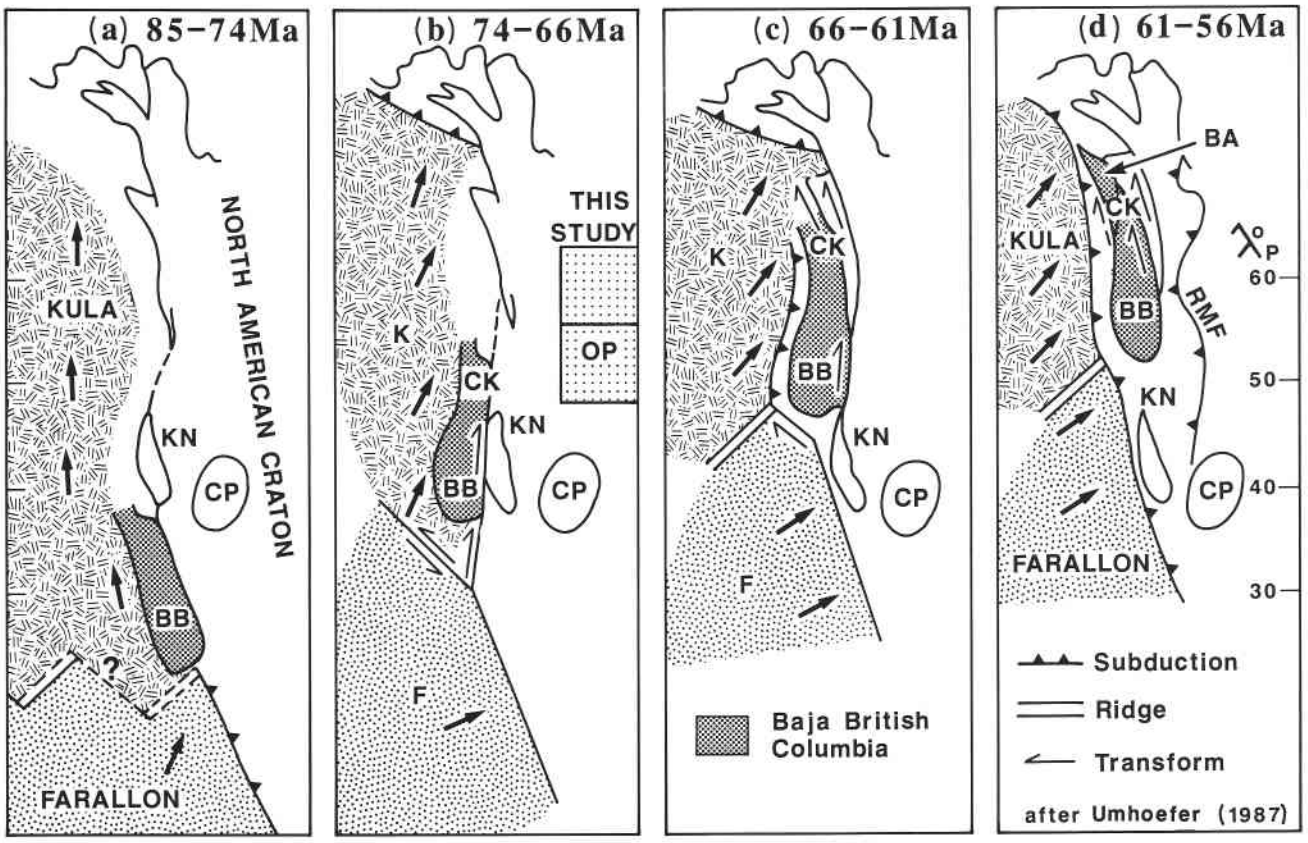


FIG. 9. Comparison of the Carmacks Group (CK) paleolatitude (OP in *b*) with Umhoefer's (1987) model for Baja British Columbia's northward translation (*d*). BA, Baja Alaska; BB, Baja British Columbia; CP, Colorado Plateau; KN, Klamath – Sierra Nevada; RMF, Rocky Mountain Front; λ_P°, paleolatitude. Ocean plate motions from Engebretson *et al.* (1985).

The trajectories also imply longitudinal motion of the Intermontane Superterrane, and this would imply that a large area of crust (approximately the stippled triangle of Fig. 8 with OP at its centre) has been destroyed between the Intermontane Belt and cratonic North America since latest Cretaceous time. There is no evidence for such post-70 Ma longitudinal convergence. Hence, it seems probable that the coupling between the Carmacks Group and the Kula plate was not perfect, specifically for the longitudinal component of the motion.

A model developed by Umhoefer (1987) accommodates this imperfect coupling. He assumed that the Kula plate motion relative to North America was almost pure transform (~100% coupling) between 85 and 66 Ma, along the north-trending margin of the North American plate. Prior to 66 Ma the Kula plate was subducting farther north. A plate reorganization at 66 Ma shifted the locus of Kula plate subduction to a more southerly latitude. This shift resulted in a dextral oblique-convergent (i.e., transpressional) margin between the Kula and North American plates west of Baja British Columbia (Fig. 9). From then on, northward motion of the Carmacks Group (CK) occurred along inboard dextral faults, resulting in a cumulative displacement of about 1000 km. Umhoefer's scenario, summarized in Fig. 9, is consistent with Gabrielse's (1985) estimate of dextral offset on the TNRMT fault system and is in excellent agreement with the displacement we have observed paleomagnetically for the Carmacks Group in the Whitehorse Trough as shown in Fig. 9*b*.

Docking time

Finally, we discuss the docking time of the Whitehorse Trough portion of Baja British Columbia. Direct evidence is absent. However, paleomagnetic data are available from the Paleocene Cantwell Formation in the central Alaska Range (Hillhouse and Grommé 1982). This formation is spatially related to, but younger than, the Carmacks Group, in that both units lie between the TNRMT fault and the Denali–Shakwak fault. Hillhouse and Grommé (1982) reported 9 ± 8° of *southward* displacement of the Cantwell Formation relative to the craton, assuming a Late Paleocene age of 56.6 ± 2.8 Ma (recalculated using revised decay constants of Steiger and Jäger 1977). With this displacement estimate being barely significant at 95% confidence, the paleolatitude of the Cantwell Formation is essentially concordant with that expected for the craton in Late Paleocene time. This would imply a docking time of about 59–54 Ma for the Whitehorse Trough, a range that is consistent with our data (Fig. 8). This is consistent with the lower limit (52 Ma) for the docking time of the southern portion of Baja British Columbia obtained from paleomagnetic data from the Eocene lavas of Kelowna (Bardoux and Irving 1989) and of Washington State (Fox and Beck 1985).

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